# Equilibrium with Complete Markets

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14 October 2026

# Roadmap

Study competitive equilibria in an economy with

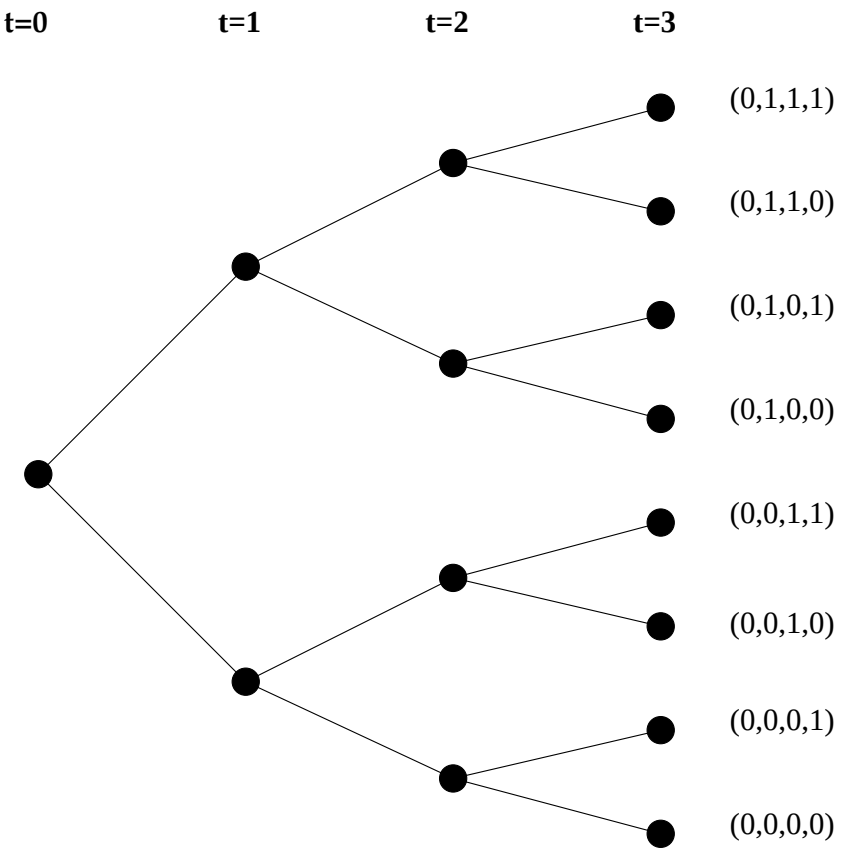
- pure exchange
- stochastic endowments
- infinite-horizon
- complete markets

Two systems of markets in the chapter:

- **Arrow–Debreu trading** (today's focus)
  - Agents meet at time 0
  - Trade claims for all possible future time periods and histories
- Sequential trading
  - Agents meet every period
  - Only trade one-period ahead state-contingent claims (Arrow securities)

# Environment: Events and Histories

- Stochastic event: $s_t \in S$
- History: $s^t = (s_0, \ldots, s_t)$
- Unconditional probability: $\pi_t(s^t)$
- Initial event is known: $\pi_0(s_0) = 1$
- $s^t$ public information



# Environment

- $I$ consumers, $i = 1, \ldots, I$
- One consumption good
- Infinite horizon, $t = 0, 1, 2, \ldots$
- Consumer $i$ receives history-dependent endowment, $y_t^i(s^t) \geq 0$
- Aggregate endowment, $Y_t(s^t) \equiv \sum_{i=1}^{I} y_t^i(s^t) > 0$
- Consumption plan $c^i = \{c_t^i(s^t)\}_{t \geq 0,\, s^t}$ specifies consumption at every date and history, e.g.

$$c^i = \left\{ \begin{array}{l} c_0^i(0), \\ c_1^i(0,0),\ c_1^i(0,1), \\ c_2^i(0,0,0),\ c_2^i(0,0,1),\ c_2^i(0,1,0),\ c_2^i(0,1,1), \\ \vdots \end{array} \right\}$$

# Preferences

The agent i utility function is

$$U^i(c^i) = \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi_t(s^t)\, u_i\left(c_t^i(s^t)\right) = E_0 \sum_{t=0}^{\infty} \beta^t u_i(c_t^i) \qquad (9.2.1)$$

- Common beliefs $\{\pi_t\}$ and common discount factor $\beta$
- $u_i$ is increasing, twice continuously differentiable, and strictly concave
- Inada condition: $\lim_{c \downarrow 0} u_i'(c) = +\infty$

# Allocations and Feasibility

- **Pure exchange:** no production or storage technology
- Feasibility requires

$$\sum_{i=1}^{I} c_t^i(s^t) \leq \sum_{i=1}^{I} y_t^i(s^t) = Y_t(s^t) \qquad \text{for every } (t, s^t) \tag{9.2.3}$$

- Increasing utility makes the resource constraint bind at an optimum
- Goods clear separately at each node

# The Social Planner

For given Pareto weights $\lambda_i > 0$, the planner solves

$$\max_{\{c^i\}_{i=1}^I} \quad \sum_{i=1}^{I} \lambda_i \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi_t(s^t)\, u_i\left(c_t^i(s^t)\right)$$

subject to $\sum_i c_t^i(s^t) \leq Y_t(s^t)$ at every node.

- Pareto optimality: no agent can gain without another losing
- Only relative weights matter, so normalize $\sum_i \lambda_i = 1$
- Weights remain fixed across dates and histories

# Planner's First-Order Conditions

Attach multiplier $\theta_t(s^t)$ to each node's resource constraint:

$$\mathcal{L} = \sum_{t=0}^{\infty} \sum_{s^t} \left[ \sum_i \lambda_i \beta^t \pi_t(s^t)\, u_i(c_t^i(s^t)) + \theta_t(s^t) \left( Y_t(s^t) - \sum_i c_t^i(s^t) \right) \right]$$

The first-order condition for $c_t^i(s^t)$ is

$$\boxed{\lambda_i \beta^t \pi_t(s^t)\, u_i'(c_t^i(s^t)) = \theta_t(s^t)} \tag{9.4.2}$$

- The planner equalizes weighted marginal utility at each node
- $\theta_t(s^t)$ is the value of an extra unit of resources there

# Efficient Risk Sharing

Taking the ratio of the first-order conditions for $i$ and $j$ gives

$$\boxed{\frac{u_i'(c_t^i(s^t))}{u_j'(c_t^j(s^t))} = \frac{\lambda_j}{\lambda_i}} \qquad \text{for all } (t, s^t) \qquad (9.4.3)$$

- Common beliefs and common discounting cancel from this ratio
- Marginal utility ratios stay constant across dates and histories
- Aggregate shocks still change the resources available for consumption

# Consumption and Aggregate Resources

For fixed weights, write consumption relative to consumer 1:

$$c_t^i(s^t) = (u_i')^{-1}\left[\frac{\lambda_1}{\lambda_i} u_1'\left(c_t^1(s^t)\right)\right] \tag{9.4.4}$$

Substitute into feasibility:

$$\sum_i (u_i')^{-1}\left[\frac{\lambda_1}{\lambda_i} u_1'\left(c_t^1(s^t)\right)\right] = Y_t(s^t) \tag{9.4.5}$$

**Proposition 1**

An efficient allocation is a function of the realized aggregate endowment and does not depend separately on either the specific history $s^t$ preceding it or the cross-section distribution of individual endowments realized at $t, s^t$: $c_t^i(s^t) = c_\tau^i(\tilde{s}^\tau)$ for $s^t$ and $\tilde{s}^\tau$ such that $\sum_j y_t^j(s^t) = \sum_j y_\tau^j(\tilde{s}^\tau)$.

# CRRA Preferences Example

Suppose all consumers have same CRRA utility function:

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma} \qquad (\gamma \neq 1)$$

Efficient risk sharing gives

$$\frac{c_t^i(s^t)}{c_t^j(s^t)} = \left(\frac{\lambda_i}{\lambda_j}\right)^{1/\gamma}$$

In this example, agents consume **constant shares** of the aggregate income

$$\boxed{c_t^i(s^t) = \alpha_i Y_t(s^t)} \qquad \alpha_i = \frac{\lambda_i^{1/\gamma}}{\sum_j \lambda_j^{1/\gamma}}$$

# Arrow–Debreu Trading

- After observing $s_0$, consumers trade at time 0
- A contingent claim delivers one unit of the good at date $t$ if history $s^t$ occurs
- **Complete markets:** a claim exists for every dated history
- $q_t(s^t)$ is that claim's price in a common unit of account
- After time 0, agents execute the promised deliveries

# The Household's Problem

Given $\{q_t(s^t)\}$, consumer $i$ chooses $c^i$ to maximize $U^i(c^i)$:

$$\max_{c^i \geq 0} \quad \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi_t(s^t)\, u_i(c_t^i(s^t))$$

subject to the lifetime budget constraint

$$\underbrace{\sum_{t=0}^{\infty} \sum_{s^t} q_t(s^t) c_t^i(s^t)}_{\text{cost of consumption plan}} \leq \underbrace{\sum_{t=0}^{\infty} \sum_{s^t} q_t(s^t) y_t^i(s^t)}_{\text{value of endowment plan}} \tag{9.5.1}$$

- Each household takes every price as given
- Increasing utility makes the budget constraint bind

# Household's First-Order Conditions

$$\max_{c^i \geq 0} \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi_t(s^t)\, u_i(c_t^i(s^t)) + \mu_i \left[ \sum_{t=0}^{\infty} \sum_{s^t} q_t(s^t) y_t^i(s^t) - \sum_{t=0}^{\infty} \sum_{s^t} q_t(s^t) c_t^i(s^t) \right]$$

Deriving FOC w.r.t. $c_t^i(s^t)$ leads to:

$$\beta^t \pi_t(s^t)\, u_i'(c_t^i(s^t)) = \mu_i q_t(s^t) \tag{9.5.4}$$

- $\mu_i$ is constant across all dates and histories
- Comparing consumers gives

$$\frac{u_i'(c_t^i(s^t))}{u_j'(c_t^j(s^t))} = \frac{\mu_i}{\mu_j} \tag{9.5.5}$$

- Equilibrium has the planner's risk-sharing condition with $\lambda_i \propto \mu_i^{-1}$

# Competitive Equilibrium

Arrow–Debreu equilibrium is a price system $\{q_t(s^t)\}$ and allocation $\{c^i\}_{i=1}^I$ s.t.:

1. **Household optimality:** every $c^i$ solves consumer $i$'s problem at those prices
2. **Market clearing:** at every node,

$$\sum_i c_t^i(s^t) = \sum_i y_t^i(s^t)$$

A competitive equilibrium allocation is a particular **Pareto optimal allocation**. In the associated planner's problem, choose Pareto weights

$$\boxed{\lambda_i = \mu_i^{-1}}$$

and the planner's shadow prices equal competitive equilibrium prices:

$$\boxed{\theta_t(s^t) = q_t(s^t)}$$

# CRRA Equilibrium Example

From the first-order conditions derived above,

$$q_t(s^t) = \frac{\beta^t \pi_t(s^t)}{\mu_i} [c_t^i(s^t)]^{-\gamma}$$

Using proportional consumption, $c_t^i(s^t) = \alpha_i Y_t(s^t)$ we obtain

$$q_t(s^t) = \mu_i^{-1} \alpha_i^{-\gamma} \beta^t \pi_t(s^t) [Y_t(s^t)]^{-\gamma}. \tag{9.6.2}$$

Taking the date-0 good as numeraire, $q_0(s_0) = \mu_i^{-1} \alpha_i^{-\gamma} [Y_0(s_0)]^{-\gamma}$ leads to

$$\boxed{q_t^0(s^t) = \beta^t \pi_t(s^t) \left( \frac{Y_t(s^t)}{Y_0(s_0)} \right)^{-\gamma}} \tag{9.6.3}$$

Household budgets give consumption shares equal to wealth shares:

$$\boxed{\alpha_i = \frac{\sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t) y_t^i(s^t)}{\sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t) Y_t(s^t)}}$$

# Constant Aggregate Endowment

Two consumers receive

$$y_t^1(s^t) = s_t, \qquad y_t^2(s^t) = 1 - s_t, \qquad s_t \in [0, 1]$$

Because $Y_t(s^t) = 1$, equilibrium consumption is constant:

$$c_t^i(s^t) = \bar{c}_i, \qquad q_t^0(s^t) = \beta^t \pi_t(s^t)$$

The budget constraint gives

$$\frac{u_i'(\bar{c}_i)}{\mu_i} \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi_t(s^t) \left[\bar{c}_i - y_t^i(s^t)\right] = 0.$$

Solving for $\bar{c}_i$,

$$\boxed{\bar{c}_i = (1 - \beta) \sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi_t(s^t)\, y_t^i(s^t)} \tag{9.6.5}$$

**Complete insurance** holds even if an individual's endowment is very volatile.

# Computing an Equilibrium

**Negishi's algorithm** converts the problem of finding a fixed point in prices into the problem of maximizing a weighted sum of utilities (easier):

1. Set $\mu_1 = 1$ and guess $\mu_2, \ldots, \mu_I > 0$
2. Solve the planner's FOCs and aggregate feasibility to obtain all $c_t^1(s^t)$
3. Recover prices from $q_t(s^t) = \beta^t \pi_t(s^t) u_1'(c_t^1(s^t))$
4. Compute budget residuals

$$B_i = \sum_{t=0}^{\infty} \sum_{s^t} q_t(s^t) \left[ c_t^i(s^t) - y_t^i(s^t) \right]$$

5. Adjust multipliers until $B_i = 0$ for $i = 2, \ldots, I$

# Asset Pricing

A **redundant asset** has a replicable payout stream $d_t(s^t)$:

$$p_0^0(s_0) = \sum_{t=0}^{\infty} \sum_{s^t} q_t^0(s^t) d_t(s^t) \tag{9.7.1}$$

A **consol** pays one unit of consumption at every date $t \geq 0$:

$$P_0^{\text{consol}} = \sum_{t=0}^{\infty} \underbrace{\sum_{s^t} q_t^0(s^t)}_{\text{price of the date-}t\text{ strip}} \tag{9.7.2}$$

At history $s^\tau$, prices are in date-$\tau$ goods; sum over its continuations:

$$p_\tau^\tau(s^\tau) = \sum_{s^{\tau+1}} q_{\tau+1}^\tau(s^{\tau+1})\, \omega_{\tau+1}(s^{\tau+1})$$

$$1 = E_\tau\left[\beta \frac{u'(c_{\tau+1})}{u'(c_\tau)} R_{\tau+1}\right] \equiv E_\tau[m_{\tau+1} R_{\tau+1}]$$

# Conclusion

- Consumption depends on aggregate endowment, not separately on the current distribution of individual endowments

- With identical CRRA utility
  - consumption shares equal wealth shares
  - prices depend only on aggregate consumption